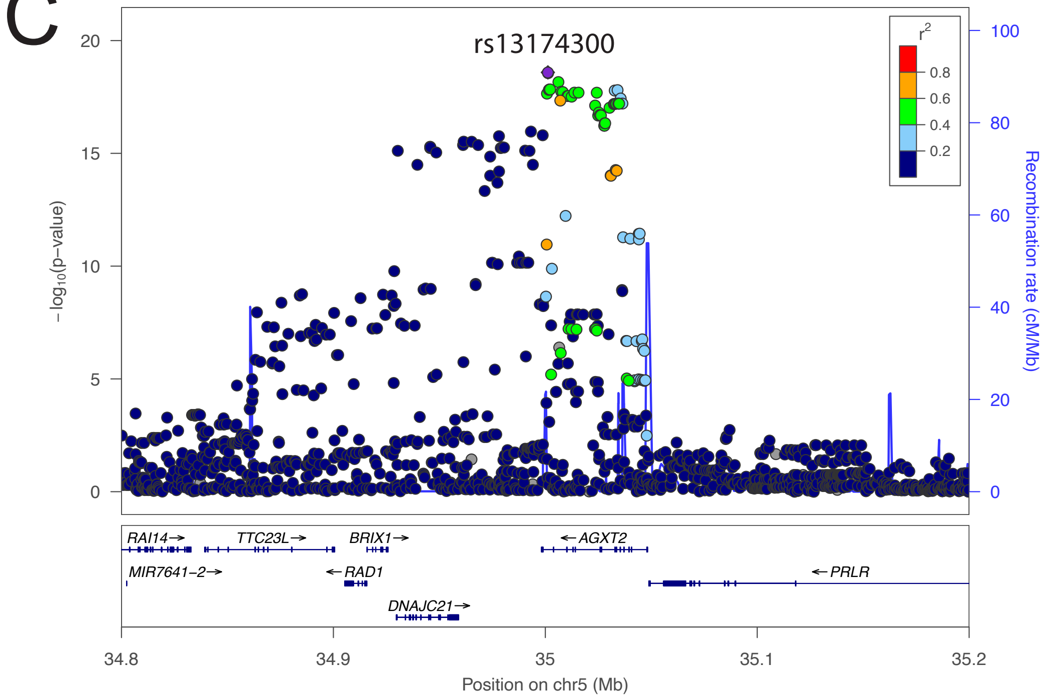
A

	FHS	N
DEMOGRAPHICS		
Total N	1613	
Female, n (%)	817 (50.7%)	1613
Age, years	35.8 ± 9.7	1613
ANTHROPOMETRICS		
BMI at Exam 5 (baseline), kg/m²	27.5 ± 4.9	1611
BMI at Exam 7 (follow-up), kg/m²	28.4 ± 5.3	1469
ΔBMI (Exam 7 − Exam 5), kg/m²	0.8 ± 2.4	1467
CARDIOMETABOLIC		
Serum creatinine, mg/dL	1.1 ± 0.2	1395
HbA1c, %	5.4 ± 0.9	925
Fasting insulin, uIU/mL	30.0 ± 12.2	1245
CLINICAL EVENTS (Exam 5 metabolomics subset)		
Type 2 diabetes, n (%)	133 (49.1%)	271
CVD history, n (%)	85 (46.4%)	183
METABOLITE RISK SCORE		
MetRS	0.16 ± 15.34	1613
METRS COMPONENT METABOLITES (inverse-normal transformed)		
Glucuronate	-0.025 ± 1.013	1613
Hippurate	0.024 ± 0.996	1613
Methyladipate/Pimelate	-0.021 ± 0.996	1613
2-Aminoisobutyric acid (BAIBA)	-0.008 ± 0.990	1613
Arginine	0.009 ± 1.006	1613
Hydroxyproline	-0.003 ± 1.010	1613
N-carbamoyl-beta-alanine	-0.004 ± 1.003	1613
Serine	0.019 ± 1.006	1613
Kynurenine	0.003 ± 0.995	1613
Malate	-0.018 ± 0.990	1613

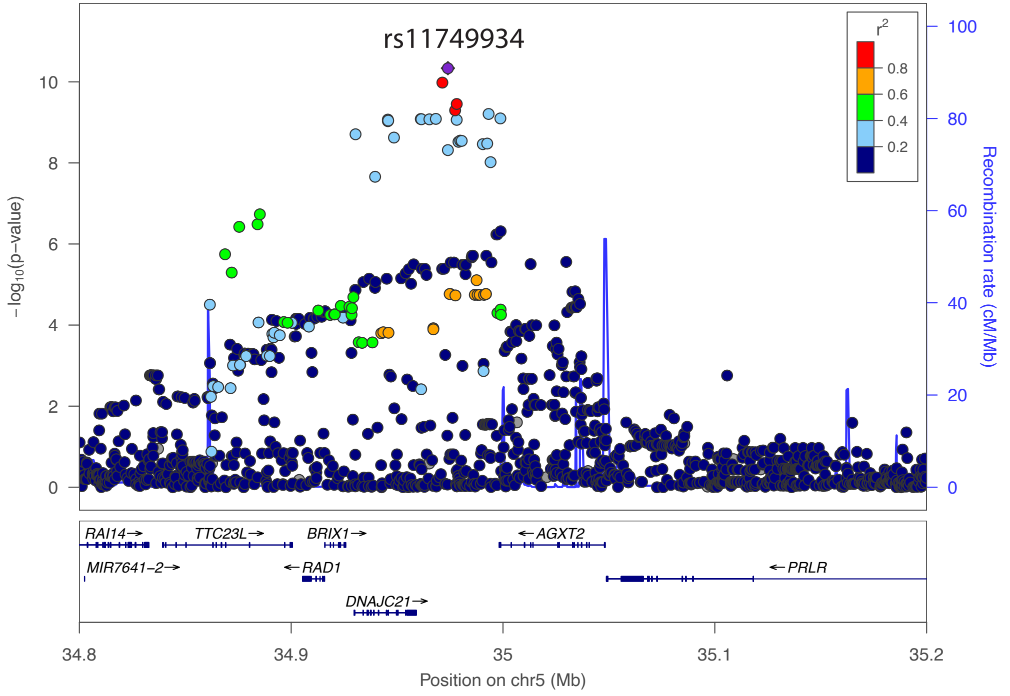
B

Metabolite	Chr	Gene	RSID	P-value
N-carbamoyl-β-alanine	chr3:161361793	Intergenic	rs182678014	2.29×10^{-8}
BAIBA	chr5:35044716	*AGXT2*	rs37376	4.77×10^{-64}
C34:2 PC	chr11:61603510	*FADS2*	rs174576	1.85×10^{-12}
Glucuronate	chr13:86072971	LINC00351	rs9594030	3.22×10^{-8}
Kynurenine	chr16:87883089	*SLC7A5*	rs2926829	1.64×10^{-8}
Hippurate	chr21:44802152	Intergenic	rs11638208	1.32×10^{-8}
MetRS	chr5:35044716	*AGXT2*	rs37376	4.88×10^{-12}

C



D



E